Data Management for Machine Learning – Assignment I

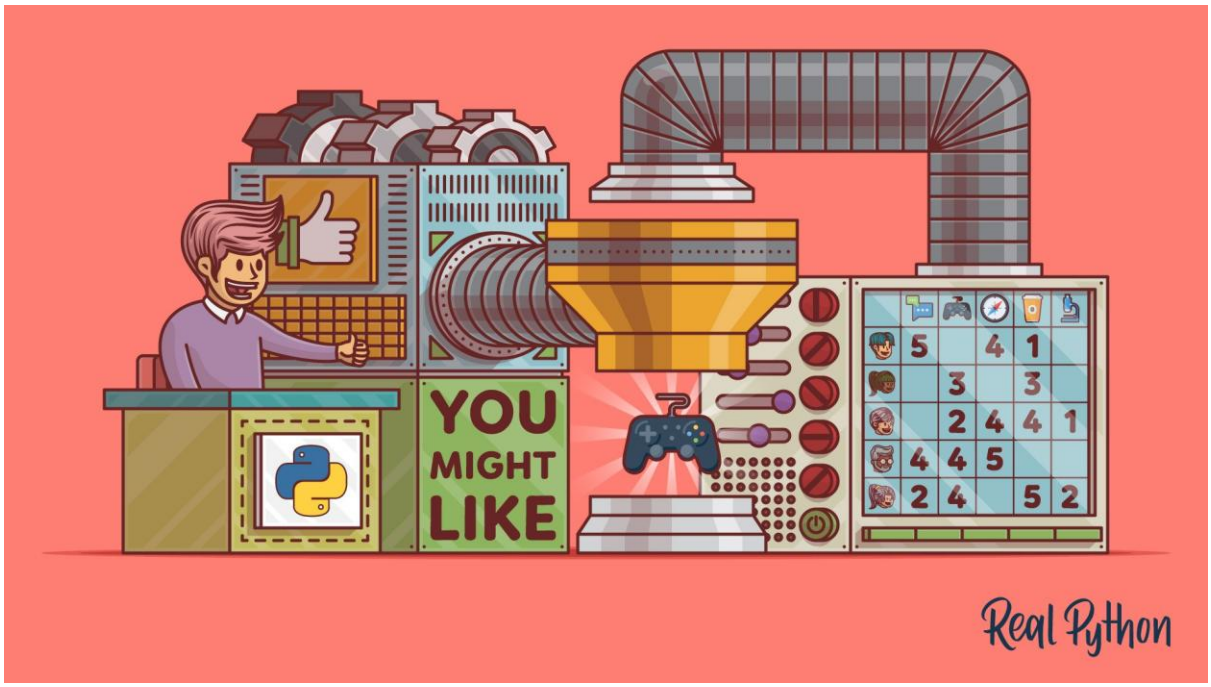
Submission Date: Refer to the assignment announcement
Weightage: 25%
Title: *End-to-End Data Management Pipeline for a Recommendation System*

Objective

Design and implement a complete data management pipeline for a recommendation system. This pipeline should support data ingestion, validation, preparation, transformation, feature management, model training, and orchestration — aligned with the practices of modern data stacks and ML workflows.

Business Context

Online recommendation systems are fundamental to modern digital platforms such as e-commerce sites, OTT services, and music streaming applications. They personalize user experiences, improve engagement, and drive sales conversions.



Your client — **RecoMart**, an e-commerce startup — wants to build a **data-driven recommendation engine** to enhance customer engagement and cross-selling opportunities. The platform collects user behavior data from multiple sources:

- Web and mobile clickstream logs
- Transactional purchase history
- Product metadata from catalogs

- External APIs (e.g., sentiment or popularity scores)

RecoMart expects a scalable and maintainable **data management pipeline** to continuously process incoming data, curate features, and train/update models for generating personalized recommendations.

Scenario

As a **Data Engineer** in RecoMart's Data Platform Team, you are tasked to design and implement an **automated, modular data pipeline** supporting both batch and near-real-time ingestion. The goal is to ensure that the recommendation model always learns from fresh, validated, and well-structured data.

Tasks

1. Problem Formulation

- Define the **business problem** clearly (e.g., product recommendation to improve conversion rate).
- Identify the **key data sources** and their attributes (e.g., users, items, ratings, transactions).
- Define the **expected outputs** from the pipeline:
    - Clean datasets for Exploratory Data Analysis (EDA)
    - Engineered features for collaborative/content-based models
    - Deployable recommendation model and inference interface
- Specify **evaluation metrics** (e.g., Precision@K, Recall@K, NDCG).

Deliverables:

- A short report (PDF) outlining problem definition, objectives, data sources, and expected pipeline outcomes.

---

2. Data Collection and Ingestion

- Ingest at least **two types of data** (e.g., user interactions from CSV files and product data from REST APIs).
- Design ingestion scripts ensuring:
    - Automated and periodic data fetching
    - Error handling and retry mechanisms
    - Logging for monitoring and audit trails

Deliverables:

- Python scripts for ingestion

- Logs showing ingestion success/failure
- Folder/bucket structure for raw data

---

3. Raw Data Storage

- Store ingested data in a **local data lake or cloud storage** (e.g., AWS S3 or local filesystem).
- Use a structured folder layout partitioned by **source**, **type**, and **timestamp**.

Deliverables:

- Storage structure documentation
- Upload scripts or configuration files

---

4. Data Profiling and Validation

- Apply validation checks to ensure **data quality and completeness**:
  - Missing values, duplicate entries, schema mismatch
  - Range and format checks (e.g., rating scale 1–5)
- Generate a **data quality report** summarizing key metrics and issues.

Deliverables:

- Python code for automated validation (using `pandas`, `great_expectations`, or `pydeequ`)
- Data Quality Report (PDF)

---

5. Data Preparation

- Perform data cleaning and preprocessing:
  - Handle missing user-item interactions
  - Encode categorical attributes (e.g., product categories, user demographics)
  - Normalize numerical variables (e.g., prices, timestamps)
- Conduct exploratory analysis showing interaction distributions, item popularity, and sparsity patterns.

Deliverables:

- Jupyter notebook or script demonstrating cleaning and EDA
- Summary plots (histograms, heatmaps)
- Prepared dataset ready for transformation

---

6. Feature Engineering and Transformation

- Create features suitable for recommendation algorithms, such as:
  - User activity frequency
  - Average rating per user/item
  - Co-occurrence or similarity-based features
- Store transformed data in a structured database or warehouse.

Deliverables:

- SQL schema
- Transformation scripts
- Summary of feature logic

---

7. Feature Store

- Implement a simple **feature store** (using Feast or a custom metadata registry).
- Document feature names, sources, and transformations.
- Enable versioned retrieval for both training and inference.

Deliverables:

- Feature store configuration/code
- Feature metadata documentation
- Sample feature retrieval demonstration

---

8. Data Versioning and Lineage

- Use tools like **DVC** or **Git LFS** to version raw and transformed data.
- Track metadata such as data source, date of ingestion, and applied transformations.

Deliverables:

- Repository structure showing dataset versions
- Documentation of versioning workflow

---

9. Model Training and Evaluation

- Train at least one recommendation model:
  - Collaborative filtering (Matrix Factorization / SVD)
  - Content-based filtering
- Evaluate model using metrics.
- Store model metadata using **MLflow** or an equivalent tracking tool.

Deliverables:

- Training and evaluation script
- Model performance report
- Tracked model metadata (run IDs, parameters, metrics)

---

10. Pipeline Orchestration

- Automate the end-to-end data pipeline using **Airflow**, **Prefect**, or **Dagster**.
- Define DAGs for data ingestion → validation → preparation → transformation → feature store → model training.
- Include logging and monitoring for failures.

Deliverables:

- Orchestration DAG/code
- Screenshots or logs from the orchestration tool showing successful execution

---

Additional Instructions

- Ensure modular code structure and clear documentation.
- Maintain logs, error handling, and reproducibility practices.
- Provide a **short demo video (5–10 mins)** explaining the pipeline flow.

Submission Requirements

- **Source Code**: Organized by pipeline stage
- **Documentation**: PDF report
- **Video Walkthrough**: End-to-end demonstration
- **Submission Format**: `.zip` file containing all deliverables

---

Evaluation Rubric (25 %)

Component	Description	Weight
Problem Formulation	Clear definition of objectives, data sources, and outputs	15%
Data Pipeline Implementation	Ingestion, storage, validation, and transformation logic	40%
Feature Store & Versioning	Effective management of features and data lineage	20%

Component	Description	Weight
Model Training & Evaluation	Performance metrics and experiment tracking	10%
Documentation & Demo	Clarity, completeness, and demonstration quality	15%
